# llmXive Automated Review of Qwen-Image-2.0-RL Technical Report

*llmXive automated review panel*

**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the support provided by citations and data within the manuscript.

**Support for Comparative Claims:** In Section 3.1, the authors assert that the pointwise reward training paradigm is "empirically superior" to pairwise training, citing Figure 1 as evidence. While the figure provides a qualitative comparison, the text lacks quantitative metrics (e.g., FID, CLIP score, or human preference win rates) comparing the two paradigms directly. The claim relies on the visual interpretation of "better visual quality" and "fewer artifacts" without statistical validation or a defined metric for "quality" in the text. To support the claim of superiority rigorously, the authors should include a table with quantitative scores or statistical significance tests for the comparison shown in Figure 1.

**Statistical Rigor of Evaluation Results:** Section 5 presents specific numerical improvements: a +2.61 increase on Qwen-Image-Bench and Elo rating gains of +78 and +93. The text describes these as "consistent gains" and "substantial improvements." However, the manuscript does not report the standard deviation, confidence intervals, or the number of independent evaluation runs used to derive these averages. In the absence of variance metrics, it is difficult to verify if these gains are statistically significant or if they could be attributed to random fluctuation in the evaluation process. The claim of "consistent" improvement is weakened without error bars or a statement on the stability of the results across multiple seeds.

**Precision of Technical Claims:** In Section 4.3, the paper states that On-Policy Distillation (OPD) "eliminates reward model dependency." This phrasing is slightly inaccurate in the context of the full pipeline. While OPD removes the need to query reward models *during the distillation phase* (the final unification step), the entire system's capability relies on the T2I and Editing teachers, which were explicitly trained using the composite reward models described in Section 3. Therefore, the system does not eliminate dependency on reward models entirely, but rather decouples the final model from the *inference-time* use of reward models. The claim should be refined to "eliminates the need for reward models during the final unification stage" to avoid misleading readers about the system's architecture.

**Citation Accuracy:** The citations for foundational works (e.g., Flow Matching, GRPO) appear accurate and correctly attributed. The citation of `liu2026gdpo` for multi-reward advantage computation is appropriate given the context of the equation. No obvious misattributions of specific algorithmic contributions were found in the Related Works section.

**Action items.**

- **[science]** In Section 3.1, the claim that pointwise training is 'empirically superior' relies on qualitative Figure 1 without quantitative metrics or statistical significance tests (e.g., p-values) in the text to support the observed difference.
- **[science]** In Section 5, specific Elo gains (+78, +93) and benchmark scores are reported without confidence intervals, standard deviations, or run counts, making the claim of 'consistent' and 'significant' improvement statistically unsupported.
- **[writing]** In Section 4.3, the claim that OPD 'eliminates reward model dependency' is misleading; it only removes dependency during the final unification stage, as the teachers still rely on reward models. Clarify to 'eliminates dependency during unification'.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure presents a qualitative comparison of two training paradigms, but the visual evidence contradicts the caption's claim that pointwise training is consistently superior, as the pairwise results appear better in the bottom row. Additionally, the column headers are ambiguous regarding whether they denote the generation method or the evaluation metric.

### Figure 2

Figure 2 effectively communicates the impact of different CFG strategies on image generation quality. The visual comparison across the three columns clearly supports the caption's claims regarding training stability and stylization preservation.

### Figure 3

Figure 3 effectively presents a qualitative comparison of three model variants across diverse T2I scenarios. The column headers clearly identify the models, and the visual progression supports the caption's claim of improved quality from Base to Mix-RL to Qwen-Image-2.0-RL (OPD).

### Figure 4

The figure content contradicts the caption's claim of showing 'portrait editing scenarios.' The examples shown are actually image generation tasks (collage creation and character design sheet generation) rather than edits of the provided input images, failing to support the caption's specific claims about editing performance.

**Action items.**

- [**science**] Figure 1: The caption claims pointwise training produces 'consistently better visual quality,' but the 'Pairwise' lighthouse image (middle row) is significantly darker and lower contrast than the 'Pointwise' version, while the 'Pairwise' doctor image (bottom row) appears sharper and better lit than the 'Pointwise' version. The visual evidence contradicts the caption's claim of consistent superiority.
- [**writing**] Figure 1: The column headers 'Pairwise Aesthetic Reward' and 'Pointwise Aesthetic Reward' are ambiguous; it is unclear if these labels refer to the training paradigm used to generate the images or the specific reward model used to evaluate them.

- **[science]** Figure 4: The caption claims to show ‘portrait editing scenarios,’ but the top row displays a collage generation task (three poses) and the bottom row displays a character design sheet (front/side/back views). Neither represents a portrait editing task (modifying an existing portrait based on instructions), making the figure content inconsistent with the caption’s stated purpose.

- **[science]** Figure 4: The top row prompt requests a ‘three-panel photographic collage,’ yet the ‘Input Image’ shows a single portrait. The generated images are not edits of the input but entirely new generations based on the prompt, which contradicts the ‘editing’ context implied by the caption and the ‘Input Image’ label.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not consistently defined at their first point of use, creating barriers for readers outside the immediate sub-field of diffusion model reinforcement learning.

In the **Abstract**, the authors introduce “OPD” (On-Policy Distillation), “T2I” (Text-to-Image), and “GRPO” (Group Relative Policy Optimization) without defining them. While “RLHF” is expanded, the other critical acronyms are not. This forces the reader to guess or search for definitions before understanding the core contributions.

In **Section 1 (Introduction)**, “VLM” (Vision-Language Model) is used in the phrase “Qwen series VLM” without expansion. Similarly, “LoRA” is mentioned in the context of existing frameworks without definition.

**Section 2 (Backgrounds)** introduces “MDP” (Markov Decision Process), “ODE” (Ordinary Differential Equation), and “SDE” (Stochastic Differential Equation). While “ODE” is partially expanded in the phrase “probability flow ordinary differential equation (ODE)”, the acronym “SDE” appears in Equation 4 without the full term “Stochastic Differential Equation” being explicitly stated in the surrounding text. “MDP” is used in Section 2.2 without definition.

**Section 3 (Reward Modeling)** mentions “Likert scale” in Section 3.1. While standard in social sciences, a brief clarification (e.g., “a 5-point rating scale”) would improve accessibility for computer science readers unfamiliar with psychometric terminology.

**Section 5 (Related Works)** uses “CoT” (Chain-of-Thought) in the phrase “chain-of-thought (CoT) reasoning”. While the expansion is present, it appears late in the sentence structure; ensuring acronyms are defined immediately upon first use is a best practice for clarity.

The paper assumes a high level of prior knowledge regarding specific RL algorithms (GRPO, Flow-GRPO) and diffusion concepts (Flow Matching, Velocity Fields) without providing a glossary or consistent expansion of acronyms. To meet the standard of a technical report accessible to a broader audience, every acronym must be defined at its first occurrence.

**Action items.**

- **[writing]** The manuscript relies heavily on domain-specific acronyms and jargon that are not consistently defined at their first point of use, creating barriers for readers outside the immediate sub-field of diffusion model reinforcement learning. In the Abstract, the authors introduce "OPD" (On-Policy Distillation), "T2I" (Text-to-Image), and "GRPO" (Group Relative Policy Optimization) without defining them. While "RLHF" is expanded, the other critical acronyms are not. This forces the reader to guess or

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent pipeline for RLHF and distillation, but several causal claims lack rigorous logical grounding or rely on unverified assumptions.

First, in Section 3.1, the argument for pointwise over pairwise training paradigms contains a logical tension. The authors claim pointwise training provides "richer supervisory signal" by encoding absolute quality, yet the implementation uses a discrete token set $\mathcal{S} = \{1, 2, 3, 4, 5\}$ with an expectation calculation. Mathematically, training a model to predict a distribution over a discrete ordinal scale is often equivalent to a pairwise ranking objective over that same scale. The paper asserts the superiority of the "absolute score" paradigm without clarifying how the discrete token implementation differs logically from a pairwise approach, or if the gain comes from the specific data annotation strategy rather than the loss function itself.

Second, the justification for the "Hybrid CFG strategy" in Section 4.1 relies on empirical observation ("leads to severe training instability") without a mechanistic explanation. The claim that applying CFG during training causes "image collapse" while omitting it causes "loss of stylization" is a strong causal assertion. However, the paper does not explain the gradient dynamics or the interaction between the conditional and unconditional branches that leads to this specific instability. Without a theoretical or ablation-based explanation of *why* the unconditional branch destabilizes the policy gradient in this specific architecture, the choice remains an unexplained heuristic.

Finally, the derivation of the On-Policy Distillation (OPD) objective in Section 4.3 and Appendix A depends critically on Assumption (A2): that the teacher velocity field is Lipschitz continuous. The paper applies this bound to RL-trained teachers. However, RL optimization often encourages sharp decision boundaries to maximize reward, which can violate Lipschitz continuity. The paper fails to address whether the RL-trained teachers satisfy this condition. If the teachers are not Lipschitz, the theoretical upper bound on the Wasserstein distance does not hold, and the logical foundation for minimizing the velocity matching loss as a proxy for distributional alignment is compromised.

**Action items.**

- [**science**] Section 3.1 claims pointwise training is superior due to 'absolute scores,' yet implements discrete token regression. This is mathematically similar to pairwise ranking over a scale. Clarify if the gain is from the loss form or data density, as the logical distinction is blurred.
- [**science**] Section 4.1 asserts CFG in training causes 'image collapse' but offers no mechanistic explanation of the gradient dynamics. The causal link between the CFG operation and this specific failure mode is asserted without derivation or theoretical support.
- [**science**] Section 4.3 derives OPD via a W2 bound assuming the teacher velocity is Lipschitz. RL-trained teachers may have sharp boundaries violating this. The paper must justify why RL teachers satisfy this assumption, or the theoretical guarantee collapses.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the superiority of its proposed methods that are not fully supported by the provided quantitative evidence, representing a moderate level of overreach.

First, in Section 3.1, the authors conclude that the pointwise reward training paradigm is "empirically superior" to pairwise training. This conclusion is drawn almost exclusively from the qualitative comparison in Figure 3, which shows generated images. While the visual difference is presented, the paper lacks quantitative validation of the reward models themselves (e.g., Spearman correlation with human judgments) or a controlled ablation study where RL is run with both reward types and compared on a standard benchmark. Claiming a fundamental methodological superiority based solely on qualitative output images is an over-extrapolation.

Second, the claim in Section 4.1 that the hybrid CFG strategy "substantially reduces computational overhead" is unsupported. The text explains the mechanism (excluding the unconditional branch from the loss) but offers no data on training time, GPU memory usage, or FLOP counts compared to a baseline that might use CFG in both stages. Without these metrics, the magnitude of the reduction is speculative.

Finally, the Conclusion and Abstract use definitive language such as "significantly improves performance" and "surpasses a mixed RL baseline." While the model does improve over its own base version, Table 1 reveals that Qwen-Image-2.0-RL (57.84) still lags behind several commercial and open baselines, including GPT Image 2 (64.69) and Nano Banana 2.0 (59.82). The narrative implies a state-of-the-art status that the data does not fully support. The authors should temper these claims to accurately reflect that the method yields consistent gains over the specific base model and a specific mixed-RL ablation, rather than implying a general dominance over the field.

**Action items.**

- [**writing**] The paper makes several strong claims regarding the superiority of its proposed methods that are not fully supported by the provided quantitative evidence, representing a moderate level of overreach. First, in Section 3.1, the authors conclude that the pointwise reward training paradigm is "empirically superior" to pairwise training. This conclusion is drawn almost exclusively from the qualitative comparison in Figure 3, which shows generated images. While the visual difference is presented, the

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a technical report on a reinforcement learning pipeline for image generation, focusing on safety and ethics through the lens of data collection, model alignment, and potential misuse.

**Human Data and Consent:** In Section 3.1 ("Reward Model Training Paradigms"), the authors describe collecting datasets where images are scored by human annotators on a 5-point Likert scale for quality and fidelity. While the methodology is described, the paper lacks a dedicated statement confirming that this human annotation process received Institutional Review Board (IRB) approval or that informed consent was obtained from the annotators. Given the involvement of human subjects in the research pipeline, explicit confirmation of ethical oversight is required to ensure compliance with standard research ethics guidelines.

**Bias and Deepfake Risks:** The paper introduces specific reward modules for "Portrait" generation (Section 3.2) and "Face identity consistency" (Section 3.3). These modules are designed to optimize for "facial attractiveness" and "identity preservation." The manuscript does not discuss the potential for these reward signals to encode or amplify societal biases regarding beauty standards (e.g., racial or gender bias). Furthermore, the capability to preserve face identity with high fidelity, combined with instruction-following editing, significantly lowers the barrier for generating deepfakes. The authors should address these dual-use risks, either by discussing the limitations of their identity preservation metrics or by outlining intended safeguards and usage policies to prevent malicious applications.

**Data Privacy and Benchmarks:** The evaluation section (Section 5) relies on "Qwen-Image-Bench" and "Q-Judger," which are trained on over 130K human-labeled image-prompt pairs. The paper does not specify the source of these images or whether they contain personally identifiable information (PII) or copyrighted material. If the benchmark includes real human faces or private data, the authors must clarify how privacy was protected during the dataset construction and annotation phases.

**Recommendation:** The paper should be revised to include a "Data Ethics and Safety" subsection or paragraph. This section must explicitly state the IRB status of human annotation, discuss the potential for bias in the portrait/aesthetic reward models, and address the dual-use risks associated with high-fidelity face identity preservation.

**Action items.**

- [**writing**] The paper describes training reward models on human-annotated datasets for aesthetics and portrait fidelity (Sec 3.1) but lacks an explicit statement regarding IRB approval, informed consent procedures, or ethical review board oversight for the human annotation process.
- [**writing**] The 'Portrait reward' (Sec 3.2) and 'Face identity consistency' (Sec 3.3) modules explicitly optimize for facial attractiveness and identity preservation. The manuscript does not address potential biases in these reward signals (e.g., Eurocentric beauty standards) or the risk of generating deepfakes, nor does it propose mitigation strategies or usage guidelines.
- [**writing**] The evaluation relies on 'Qwen-Image-Bench' and 'Q-Judger' (Sec 5), which are described as trained on human-labeled data. The paper must clarify the provenance of this data, specifically whether it includes personally identifiable information (PII) or copyrighted images, and how privacy was maintained during the creation of the benchmark.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling pipeline for RLHF and distillation in image generation, but the scientific evidence supporting the specific methodological choices and performance claims requires quantitative reinforcement.

First, the central claim in Section 3.1 that pointwise reward training outperforms pairwise training is supported only by qualitative visual examples in Figure 2. While the authors argue that absolute scores provide richer signals, they do not provide statistical evidence. The review requires the authors to report quantitative metrics, such as the Spearman correlation or Kendall's tau between the reward model scores and human preference rankings on a held-out validation set, to empirically validate that the pointwise model is indeed a better proxy for human judgment.

Second, the evaluation section relies heavily on Elo ratings (Section 5, Figure 3) and automated benchmark scores (Table 1). The reported Elo gains (+78 and +93) are substantial, but the paper omits the number of battles conducted and any measure of statistical significance (e.g., confidence intervals or p-values). In arena settings, variance can be high; without reporting the sample size and statistical significance, it is difficult to rule out that these gains are due to random noise. Similarly, the Qwen-Image-Bench results depend entirely on the "Q-Judger" model. The paper fails to report the correlation between Q-Judger's scores and human annotators on a gold-standard test set. Without establishing the validity of the automated judge, the benchmark scores are not robust evidence of improvement.

Finally, the comparison between the proposed On-Policy Distillation (OPD) and the "Mix-RL" baseline (Section 4.3) is presented through qualitative figures (Figures 5 and 6). While the visual differences appear clear, the claim that OPD avoids "cross-task optimization conflicts" needs quantitative backing. The authors should provide numerical metrics (e.g., FID, CLIP score, or specific task success rates) comparing the two approaches on a standardized test set to demonstrate that the decomposed strategy yields statistically significant improvements over joint training.

**Action items.**

- [**science**] The paper claims pointwise reward training is superior to pairwise (Sec 3.1, Fig 2) but lacks quantitative metrics (e.g., correlation coefficients, MSE) comparing the reward models' alignment with human judgments. Provide statistical evidence that the pointwise model's scores correlate significantly better with human preferences than the pairwise model's scores.
- [**science**] The Elo rating improvements (+78 T2I, +93 Edit) are presented without confidence intervals or significance testing (Sec 5). Given the variance inherent in arena battles, report the number of battles, standard errors, or p-values to confirm these gains are statistically significant and not due to random fluctuation.
- [**science**] The 'Qwen-Image-Bench' results (Tab 1) rely on 'Q-Judger,' a model trained on 130K pairs. The paper does not report the inter-rater reliability (e.g., Cohen's kappa) between Q-Judger and human annotators on a held-out test set. Without this validation, the automated benchmark scores are not robust evidence of human preference alignment.
- [**science**] The comparison between the proposed OPD pipeline and the 'Mix-RL' baseline (Sec 4.3, Figs 5-6) is purely qualitative. To support the claim that OPD avoids optimization conflicts, provide quantitative metrics (e.g., FID, CLIP score, or specific task accuracy) comparing the two methods on a standardized test set.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_statistical_analysis.md`

The paper presents a comprehensive pipeline for RLHF and distillation in image generation, but the statistical rigor of the evaluation and ablation studies requires strengthening to support the quantitative claims made.

First, in **Section 4 (Evaluation)**, the authors report precise point estimates for benchmark performance (e.g., Qwen-Image-Bench overall score of 57.84) and Elo ratings (1193 for T2I, 1349 for editing). However, the manuscript lacks any measure of uncertainty, such as confidence intervals, standard deviations, or standard errors. Given that diffusion model generation is stochastic and human preference voting (Elo) involves sampling variance, reporting only point estimates is insufficient. The authors should report the results of multiple independent runs (e.g., mean $\pm$ std dev) or provide confidence intervals derived from bootstrapping to demonstrate that the observed improvements (e.g., +2.61 points, +78 Elo) are statistically significant and not artifacts of random variation.

Second, the comparison between **pointwise and pairwise reward training paradigms in Section 3.1** relies heavily on qualitative visual evidence (Figure 2) and a textual assertion that pointwise is "empirically superior." To substantiate this claim, the authors should provide quantitative metrics comparing the two reward models. This could include the mean and variance of the reward scores assigned to a held-out test set, or a statistical test (e.g., paired t-test) on the downstream RL performance when using each reward model. Without this, the preference for the pointwise paradigm remains an anecdotal observation rather than a statistically validated finding.

Third, regarding the **multi-reward advantage computation (Eq. 10)**, the method relies on group-level normalization ($\mu_k, \sigma_k$). The statistical stability of these estimates depends heavily on the group size $G$. The paper does not explicitly state the value of $G$ used during training or discuss the potential bias and variance introduced if $G$ is small. A sensitivity analysis showing how the choice of $G$ affects the advantage distribution and final model performance would strengthen the reproducibility and robustness of the proposed training framework.

Finally, while the **On-Policy Distillation (OPD)** derivation in the Appendix is mathematically sound, the empirical validation of the distillation process (Section 3.3) lacks statistical comparison against the Mix-RL baseline. The claim that OPD "consistently outperforms" Mix-RL is supported by qualitative figures and single-point metrics. A statistical test comparing the distributions of scores or Elo ratings across multiple seeds would provide stronger evidence for the superiority of the decomposed strategy.

**Action items.**

- [**science**] In Section 4 (Evaluation), the paper reports specific point estimates for Qwen-Image-Bench scores (e.g., 57.84) and Elo ratings (e.g., 1193) without providing confidence intervals, standard errors, or p-values. Given the stochastic nature of diffusion sampling and human preference voting, statistical significance testing or uncertainty quantification is required to validate that the reported gains (+2.61, +78 Elo) are not due to random variance.
- [**science**] Section 3.1 compares pointwise vs. pairwise reward training paradigms. The text claims the pointwise approach is 'empirically superior' based on qualitative figures, but lacks quantitative statistical evidence (e.g., mean reward scores with variance, t-tests, or effect sizes) to support the claim that the difference is statistically significant rather than anecdotal.
- [**science**] The multi-reward advantage computation in Eq. 10 uses per-prompt-group normalization. The paper does not specify the group size (G) used for calculating the mean and standard deviation, nor does it discuss the stability of these statistics for small group sizes. Sensitivity analysis or justification for the chosen G is needed to ensure the advantage estimates are robust.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical depth and generally maintains a clear, professional academic tone. The logical flow between sections is strong, particularly in the transition from the problem statement in the Introduction to the specific contributions and the detailed methodology. The use of structured paragraphs and clear signposting (e.g., "In this work, we address these challenges...") aids readability significantly.

However, there are several minor but noticeable grammatical errors and typos that detract from the polish of the writing. In Section 5 (Related Works), the sentence "Our work builds upon these methods, introduce a hybrid CFG strategy..." suffers from a subject-verb agreement error; "introduce" should be "introduces" or the sentence structure should be adjusted to "and introduces." Similarly, in Section 3.1, the phrasing "We collect images datasets" and "We collect image pairs dataset" is awkward and grammatically incorrect; these should be revised to "image datasets" and "an image pairs dataset" (or similar) for proper noun usage and article agreement.

Additionally, a likely typo exists in the Conclusion (Section 6), where "TI2I tasks" is mentioned. Given the consistent use of "T2I" throughout the document, this is almost certainly a typo that should be corrected to maintain consistency. While these issues do not obscure the scientific meaning, addressing them is necessary to meet the standard of a polished technical report. The rest of the prose is concise and effective, with well-constructed sentences that clearly explain complex RL and diffusion concepts.

**Action items.**

- [**writing**] In Section 5 (Related Works), the sentence 'Our work builds upon these methods, introduce a hybrid CFG strategy...' contains a grammatical error. The verb 'introduce' should be 'introduces' to agree with the singular subject 'Our work', or the clause should be rephrased (e.g., '...and introduces...').
- [**writing**] In Section 3.1 (Reward Model Training Paradigms), the phrase 'We collect images datasets' is grammatically incorrect. It should be 'We collect image datasets' (using 'image' as a singular attributive noun) or 'We collect datasets of images'.
- [**writing**] In Section 3.1, the phrase 'We collect image pairs dataset' lacks an article and proper number agreement. It should be corrected to 'We collect an image pairs dataset' or 'We collect image pairs datasets'.
- [**writing**] In Section 6 (Conclusion), the first contribution lists 'TI2I tasks'. This appears to be a typo for 'T2I' (Text-to-Image), which is the standard abbreviation used throughout the rest of the paper. Please verify and correct.